UNIT 3 · TOPIC 2(A)

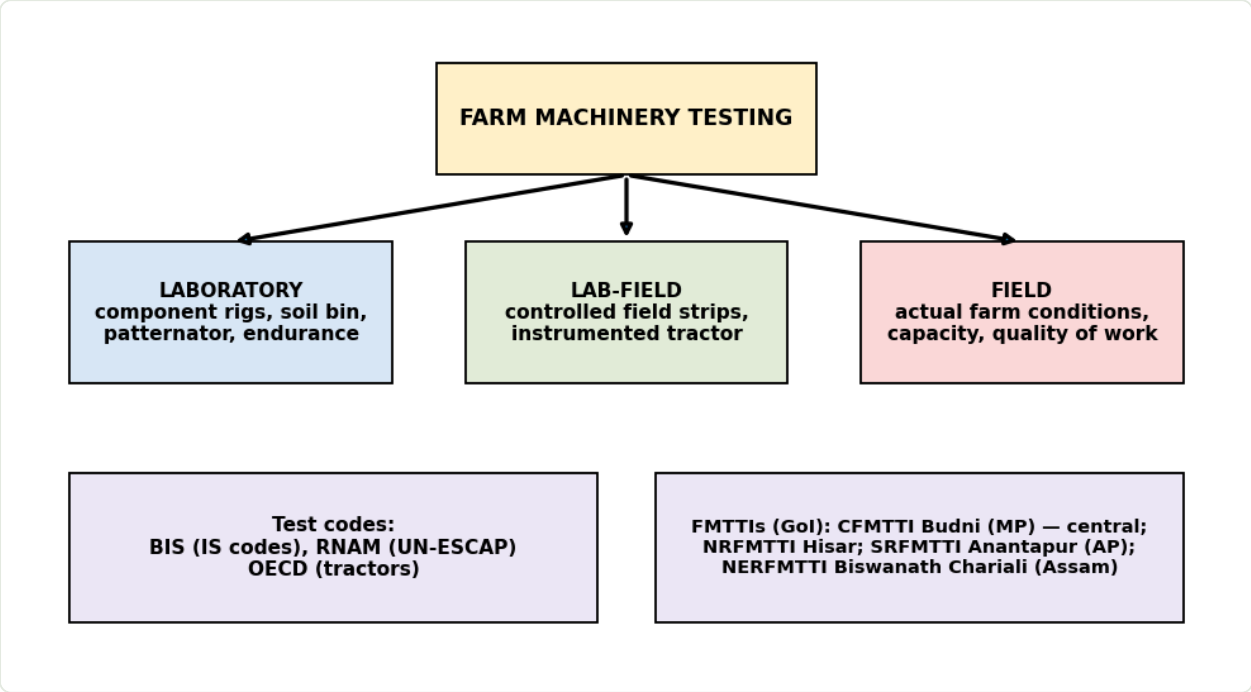
Testing Framework, Test Codes and Organisations

Syllabus topic: Methods of testing and performance evaluation of tillage equipment, seed drills and planters

Why test

- Verify manufacturer claims.
- Ensure minimum performance and safety for certification and **subsidy eligibility**.
- Generate comparable data for buyers.
- Feed design improvement.

A test evaluates: specifications, materials, performance (quality + rate of work), power/energy, ease of operation & adjustment, safety, and durability (endurance run + breakdown record).



Levels of farm machinery testing, test codes and FMTTIs

Levels and types of test

Level	What it involves
Laboratory	Controlled — test rigs, soil bins, patternators, endurance runs
Laboratory-field	Instrumented machine on selected field strips
Field	Actual operating conditions over a season

Tests may be **initial/commercial** (full, for market release) or **batch/confidential** (quality check, developmental).

Test codes

- **RNAM** (Regional Network for Agricultural Machinery, UN-ESCAP) codes for seed drills/planters, sprayers, threshers etc. — widely adopted across Asia.
- **BIS** Indian Standards — e.g., **IS 6316**, the test code for seed-cum-fertilizer drills.
- **OECD** codes for tractors (Unit 4).

Testing institutes in India

FMTTI	Location
CFMTTI — the central institute (also tractor testing)	Budni (MP)
NRFMTTI	Hisar (Haryana)
SRFMTTI	Anantapur/Garladinne (AP)
NERFMTTI	Biswanath Chariali (Assam)

ICAR-CIAE Bhopal and SAUs also test prototypes.

The test report

Specifications & checking; the results of each test with methods and instruments; defects/breakdowns; a summary assessment — the basis of the machine's test certificate.

Practice MCQs

Q36. Farm machinery testing that uses soil bins and test rigs under controlled conditions is:

(a) Field test
(b) Laboratory test
(c) Batch test
(d) Endurance test

Q37. RNAM test codes were developed under:

(a) FAO
(b) UN-ESCAP Regional Network for Agricultural Machinery
(c) WTO
(d) ISO

Q38. The Indian Standard test code for seed-cum-fertilizer drills is:

(a) IS 6316
(b) IS 9020
(c) IS 4468
(d) IS 12207

Q39. Tractor test codes accepted internationally are published by:

(a) RNAM
(b) OECD
(c) WHO
(d) BIS only

Q40. The Central Farm Machinery Training & Testing Institute is located at:

(a) Hisar
(b) Budni (MP)
(c) Anantapur
(d) Bhopal

Q41. The northern region FMTTI is at:

(a) Ludhiana
(b) Hisar
(c) Pantnagar
(d) Karnal

Q70. An endurance (durability) test primarily records:

(a) Paint gloss

(b) Breakdowns, wear and defects over long running hours

(c) Seed rate

(d) Droplet size

Answers

Q36 — (b) Laboratory test

Q37 — (b) UN-ESCAP Regional Network for Agricultural Machinery

Q38 — (a) IS 6316

Q39 — (b) OECD

Q40 — (b) Budni (MP)

Q41 — (b) Hisar

Q70 — (b) Breakdowns, wear and defects over long running hours

Sources: RNAM (UN-ESCAP) Test Codes and Procedures for Farm Machinery; BIS Indian Standards for seed drills, sprayers and dusters; Kepner et al. (performance concepts); Jagdishwar Sahay; FMTTI (Budni) test procedures; NPTEL lectures.